

WEAK-EXCEDANCE EXTRACTION: EXACT IMAGES, FERRERS FIBRES, AND FIBONACCI RIGHT-INVERSE TOWERS

ANONYMOUS

ABSTRACT. For a permutation $\pi = \pi_1 \cdots \pi_n$, retain the letters satisfying $\pi_i \geq i$ and standardize them in their original order. We study the resulting rank-changing permutation map. If $\sigma \in \mathfrak{S}_m$ and $d(\sigma) = \max_i (i - \sigma_i)$, then σ occurs from source rank n if and only if $n \geq m + d(\sigma)$; a high-shift/low-tail section realizes every admissible rank. We also give an exact fibre formula for every target as a sum of deficient Ferrers-board completion counts. The only recurrent states are identities. On nonidentity targets, iterating the minimum-rank section produces an inverse ray whose rank and maximum drop evolve by $(m, d) \mapsto (m + d, m)$, hence by Fibonacci matrix powers. This is a local one-step minimality theorem, not a global minimum theorem for iterated preimages and not a maximum absorption clock. Classical weak-excedance, bounded-drop, tableau, Bruhat, and Bell-enumeration results are explicitly subtracted. An exact audit executes 3,689,489 assertions, including explicit rank-boundary tests and a counterexample to a withdrawn pointwise clock claim.

1. MAP, SCOPE, AND MAIN THEOREM

Let $\mathfrak{S}_n$ be the permutations of $[n]$ in one-line notation. If v is a word of distinct integers, write $\text{std}(v)$ for its standardization. Define

$$(1) \quad W(\pi) = \text{std}(\pi_i : \pi_i \geq i), \quad \pi \in \mathfrak{S}_n,$$

where the retained letters stay in their original order. Position one is always retained, so (1) is a self-map of $\mathfrak{S}_{\leq N} = \bigsqcup_{1 \leq n \leq N} \mathfrak{S}_n$ for every $N \geq 1$. Put

$$(2) \quad d(\sigma) = \max_{1 \leq i \leq m} (i - \sigma_i), \quad \sigma \in \mathfrak{S}_m.$$

This is the classical maximum-drop statistic.

The static ownership boundary is deliberately broad. Excedance-set enumeration is due to Ehrenborg–Steingrímsson [7]; bounded maximum drop and its enumerators are treated by Chung–Claesson–Dukes–Graham [6] and Chen–Chen [5]; permutation-tableau structure is developed by Steingrímsson–Williams [8]; and Bergeron–Gagnon organize weak-excedance position/value classes through a Bruhat quotient [3]. All of those static statistics, distributions, and structures receive zero contribution credit here.

There is also an exact collision at the identity aggregate. Beyene–Backelin–Mantaci–Fufa, using the transposition-array interface of Baril [1, 2], prove that permutations whose weak-excedance-letter subword is increasing are Bell-number enumerated [4, Theorem 27]. In the notation below this owns

$$(3) \quad \sum_{m=1}^n |W_n^{-1}(\text{id}_m)| = B_n.$$

Equation (3) and its transposition-array proof are not claimed. Our fibre formula retains both the target permutation and its rank, including nonidentity targets. Generic Ferrers-board matching is likewise background; the residual is the conjunction of the literal extraction map, its exact target obstruction and sections, its target-resolved fibres, and the dynamics of one canonical right inverse. A bounded exact-map search non-hit is not evidence of novelty, priority, or clearance.

For $n \geq m$, write $h = n - m$. For $A = \{a_1 < \cdots < a_m\}$ and $P = \{p_1 < \cdots < p_m\}$ contained in $[n]$, call (A, P) σ -admissible if

$$(4) \quad p_i \leq a_{\sigma_i} \quad (1 \leq i \leq m).$$

Let $B = [n] \setminus A$ and $Q = [n] \setminus P = \{q_1 < \dots < q_h\}$, and put

$$(5) \quad K(B, Q) = \prod_{j=1}^h (|\{b \in B : b < q_j\}| - (j-1)),$$

interpreted as zero if any factor is nonpositive; when $h = 0$, the empty product is 1.

Theorem 1 (images, fibres, recurrence, and inverse tower). *Let $\sigma \in \mathfrak{S}_m$.*

(1) *For every $n \geq m$,*

$$(6) \quad \sigma \in \mathcal{W}(\mathfrak{S}_n) \iff n \geq m + d(\sigma).$$

Whenever $h = n - m \geq d(\sigma)$, the explicit right section is

$$(7) \quad R_n(\sigma) = (\sigma_1 + h, \dots, \sigma_m + h, 1, \dots, h).$$

Thus $m + d(\sigma)$ is the exact minimum source rank.

(2) *For every source rank $n \geq 1$,*

$$(8) \quad |\mathcal{W}_n^{-1}(\sigma)| = \begin{cases} 0, & 1 \leq n < m, \\ \sum_{(A,P) \text{ } \sigma\text{-admissible}} K([n] \setminus A, [n] \setminus P), & n \geq m. \end{cases}$$

This includes zero fibres. At the same rank $n = m$, the fibre has size one for $\sigma = \text{id}_m$ and zero otherwise.

(3) $\mathcal{W}(\pi) = \pi$ if and only if π is an identity. Every nonidentity step strictly lowers rank; hence the recurrent states of $\mathfrak{S}_{\leq N}$ are $\text{id}_1, \dots, \text{id}_N$.

(4) Suppose σ is nonidentity, put $d = d(\sigma) > 0$, and define $\sigma^{(0)} = \sigma$ and

$$(9) \quad \sigma^{(t+1)} = R_{m_t+d_t}(\sigma^{(t)}), \quad m_t = |\sigma^{(t)}|, \quad d_t = d(\sigma^{(t)}).$$

Then

$$(10) \quad (m_{t+1}, d_{t+1}) = (m_t + d_t, m_t), \quad \mathcal{W}(\sigma^{(t+1)}) = \sigma^{(t)}.$$

With $F_0 = 0, F_1 = 1$, for every $t \geq 1$,

$$(11) \quad m_t = F_{t+1}m + F_t d, \quad d_t = F_t m + F_{t-1} d.$$

Every edge of this inverse ray has minimum possible one-step source rank. If τ is the first hitting time of an identity, then

$$(12) \quad \tau(\sigma^{(t)}) = \tau(\sigma) + t.$$

Part 4 says *locally minimum*: each individual lift uses a minimum-rank preimage of its immediate target. It does not assert that the composite tower minimizes rank among all t -step preimages.

2. IMAGES AND DEFICIENT COMPLETIONS

Proof of Theorem 1(1). Suppose $\pi \in \mathfrak{S}_n$ maps to $\sigma \in \mathfrak{S}_m$. Let $P = \{p_1 < \dots < p_m\}$ be its selected positions, and let $A = \{a_1 < \dots < a_m\}$ be its selected values. Standardization forces the entry at p_i to be a_{σ_i} . Selection gives

$$(13) \quad i \leq p_i \leq a_{\sigma_i}.$$

Since only $h = n - m$ values lie outside A , the j th selected value satisfies $a_j \leq h + j$. Applying this to (13) yields

$$i \leq h + \sigma_i, \quad i - \sigma_i \leq h.$$

Taking the maximum proves $d(\sigma) \leq h$, hence necessity in (6).

Conversely, suppose $h \geq d(\sigma)$ and use (7). Its first m entries satisfy $\sigma_i + h \geq i$ and are therefore selected. Its final entry j occurs at position $m + j > j$ and is deficient. Hence the selected word is exactly the shifted copy $(\sigma_1 + h, \dots, \sigma_m + h)$, whose standardization is σ . This proves the section identity and the sharp threshold. $\square$

Proof of Theorem 1(2). If $n < m$, a rank- n source cannot produce a rank- m target, so the fibre is empty. Assume henceforth that $n \geq m$. Fix selected sets A and P . The selected assignment is forced: p_i receives a_{σ_i} . It consists of weak excedances exactly when (4) holds. It remains to biject the complement values B onto the complement positions Q so that each value assigned at q_j is strictly smaller than q_j .

Process $q_1, \dots, q_h$ increasingly. At q_j there are $|\{b \in B : b < q_j\}|$ eligible complement values. Every one of the $j-1$ previously assigned values is among them, because it was assigned at an earlier position $q_k < q_j$ and is smaller than q_k . Therefore the number of choices at step j is precisely the j th factor of (5). The product counts all deficient complement assignments, with a nonpositive factor correctly giving zero.

Every source determines one unique pair (A, P) and one such complement assignment. Conversely, the forced selected assignment and any deficient completion give a unique source whose selected standardized word is σ . The classes are disjoint, so summing proves (8). When $n = m$, both complements are empty and $K(\emptyset, \emptyset) = 1$. The sole pair $A = P = [m]$ is admissible exactly when $i \leq \sigma_i$ for every i , which by equality of coordinate sums holds exactly for $\sigma = \text{id}_m$. This proves the same-rank boundary. $\square$

Formula (8) is a Ferrers-board product in elementary form: at each increasing complement position, it counts the unused values lying strictly below the diagonal. Summing it over identity targets recovers the owned Bell aggregate (3); that consistency check does not transfer ownership of the rank- and target-resolved formula.

3. FORWARD ABSORPTION AND THE CANONICAL INVERSE RAY

Proof of Theorem 1(3). If $W(\pi)$ has the same rank as $\pi \in \mathfrak{S}_n$, every position was selected, so $\pi_i \geq i$ for all i . But $\sum_i \pi_i = \sum_i i$; hence equality holds term by term and $\pi = \text{id}_n$. Identities are fixed. Thus every nonidentity step strictly lowers positive integer rank, proving both absorption and the recurrent classification. $\square$

For a nonidentity σ , $d(\sigma) > 0$: otherwise $\sigma_i \geq i$ for every i , and the preceding sum argument would make σ an identity. The minimum-rank section in (9) is therefore a genuine rank increase.

Proof of Theorem 1(4). Apply Part 1 with $h = d_t$. It gives the right-section identity in (10) and proves that $m_t + d_t$ is the minimum rank of any one-step preimage of $\sigma^{(t)}$.

The first m_t entries of the section have the form $\sigma_i^{(t)} + d_t$. Their drop is

$$i - (\sigma_i^{(t)} + d_t) \leq d_t - d_t = 0.$$

The final low value j occurs at position $m_t + j$ and has drop exactly m_t . Consequently $d_{t+1} = m_t$, while the rank is visibly $m_{t+1} = m_t + d_t$. This proves the resource update.

Writing the update as multiplication by $\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$ and using its standard Fibonacci powers proves (11) by induction. Finally, the lifted state makes one step to its immediate target and is nonrecurrent; therefore each lift increases the identity hitting time by exactly one, which proves (12). $\square$

For example, starting from 21 gives resource pairs

$$(m_t, d_t) = (2, 1), (3, 2), (5, 3), (8, 5), (13, 8), \dots$$

The Fibonacci sequence here describes a chosen backward dynamical system. It does not by itself bound all forward orbits at a fixed rank.

4. EXACT CONTROL, EXCLUSIONS, AND DECLARATIONS

The accompanying deterministic verifier enumerates all 409,113 permutations through rank nine, reconstructs the literal functional graph, and checks the image theorem in 99,451 target/rank cells. It compares (8) with literal predecessor counts for all 6,985 target cells through source rank seven, then separately checks 316,646 cells with $n < m$ and all 46,233 same-rank cells through target rank eight. It also recovers the owned Bell aggregate. Finally, it takes every one of the 46,225 nonidentity targets through rank eight through six exact section lifts and checks (10)–(12). The frozen transcript records 3,689,489 exact assertions and ends in **PASS**. These computations are falsification pressure, not proofs of the all-rank statements.

Limitations and withdrawn clock claims. No maximum absorption clock is claimed. In particular, the tempting pointwise inequality

$$\tau(W(\pi)) \leq \max_{\rho \in \mathfrak{S}_{d(\pi)}} \tau(\rho)$$

is false: for

$$\pi = (11, 10, 9, 4, 1, 2, 3, 8, 5, 6, 7)$$

one has $d(\pi) = 4$, while $W(\pi) = (5, 4, 3, 1, 2)$ has tail three and the maximum tail in $\mathfrak{S}_4$ is two. The verifier reproduces this counterexample. The paper also makes no global minimum-rank claim for t -step preimages and does not infer a fixed-rank maximum from the Fibonacci inverse ray. The broad carrier pattern “extract a subsequence and standardize” receives zero credit; the diagonal predicate, maximum-drop image obstruction, and deficient completion are the specific mechanisms used here.

Data availability. The exact-control source and frozen transcript accompany the internal artifact. They use no external data, runtime network access, random sampling, or nonstandard Python package.

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